



MAX ChIP-Seq PROJECT

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MAX: MYC associated factor X

Family: belongs to the conserved basic helix-loop-helix leucine zipper (bHLHZ) family of transcription factors.

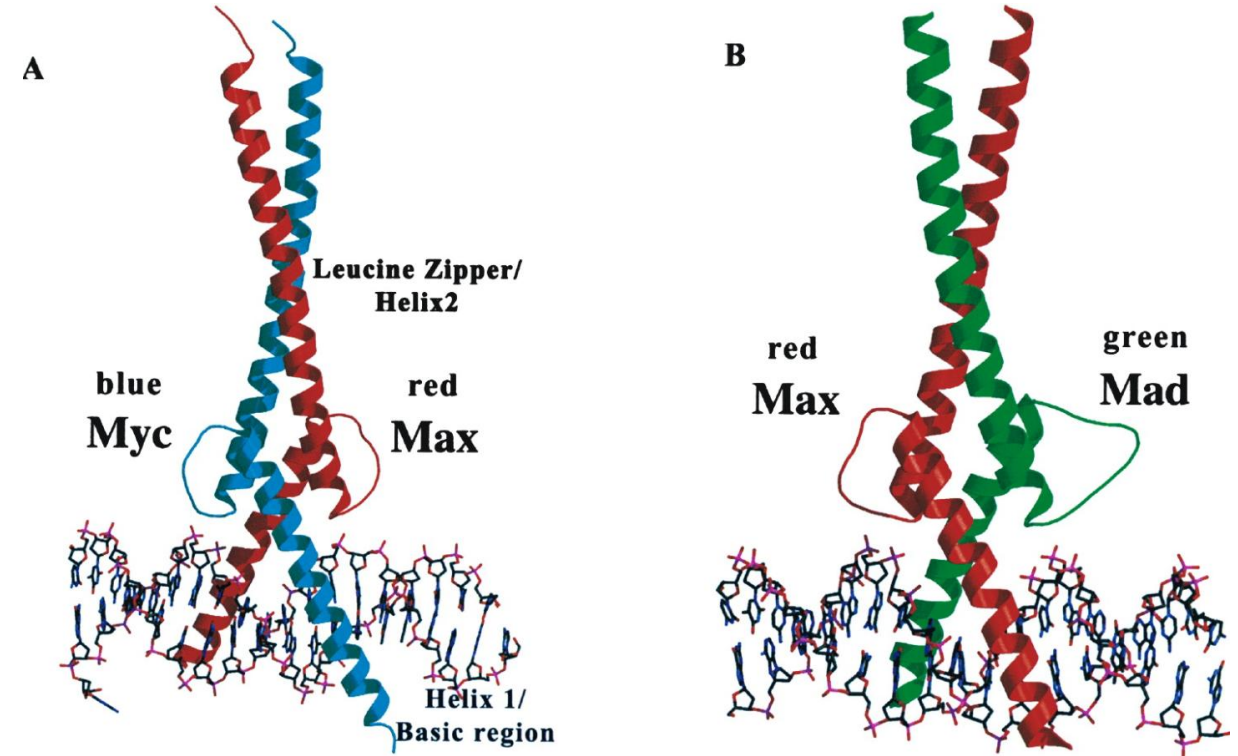
Functions:

- Acts as a central regulator of cell fate and proliferation.
- Forms heterodimers with MYC to activate transcription of growth-related genes.
- Partners with MAD proteins to repress transcription, maintaining homeostatic balance between cell division and differentiation.

DNA Binding Motif: A core hexanucleotide element, the Enhancer box (E-box,) that activate transcription at target promoters.



MAX binding motif



X-Ray Structures of Myc-Max and Mad-Max Recognizing DNA

Mapping & Quality Controls

Library	Total Reads	Uniquely Mapped	Multimapping	Non-mapping
Replicate 1	43,802,638	38,393,205 (87.65%)	1,553,555 (3.55%)	3,855,878 (8.80%)
Replicate 2	51,302,620	45,644,909 (88.97%)	1,245,170 (2.43%)	4,412,541 (8.60%)
Input (Control)	35,636,644	26,856,763 (75.36%)	2,426,649 (6.81%)	6,353,232 (17.83%)

Quality controls were successfully passed. Uniquely mapped reads for both replicates well exceed the required standard threshold, demonstrating high-quality sequencing and alignment.

Peak Calling: MACS2

CRISPR-edited datasets: q-value threshold set to 0.01

MACS2 Run	Peaks Found	Redundant Rate	Fragment Size (d)
Replicate 1	39,775	0.20 (20%)	160 bp
Replicate 2	26,726	0.14 (14%)	189 bp
Merged (Pooled)	26,308	0.15 (15%)	173 bp

Redundancy Rate below 20% are acceptable, indicating good library complexity and no severe PCR bottleneck artifacts.

Quality Controls
fully satisfied

No errors or warnings reported by MACS2

Fragment size d should typically fall between 100 bp and 200 bp. All estimated sizes are within the thresholds.

Peak Overlaps

~**87%** of Rep 2 peaks are contained in Rep 1

Intersection between replicates	Peaks in common (simple overlap)	Peaks in common (summit proximmmity)
Rep 1 vs Rep2	23307	19767

A comparison with the merge...

Intersection with merge	Peaks in common (basic overlap)	% peaks in common
Rep1 vs Merge	24944	62.7%
Rep2 vs Merge	23007	86.1%

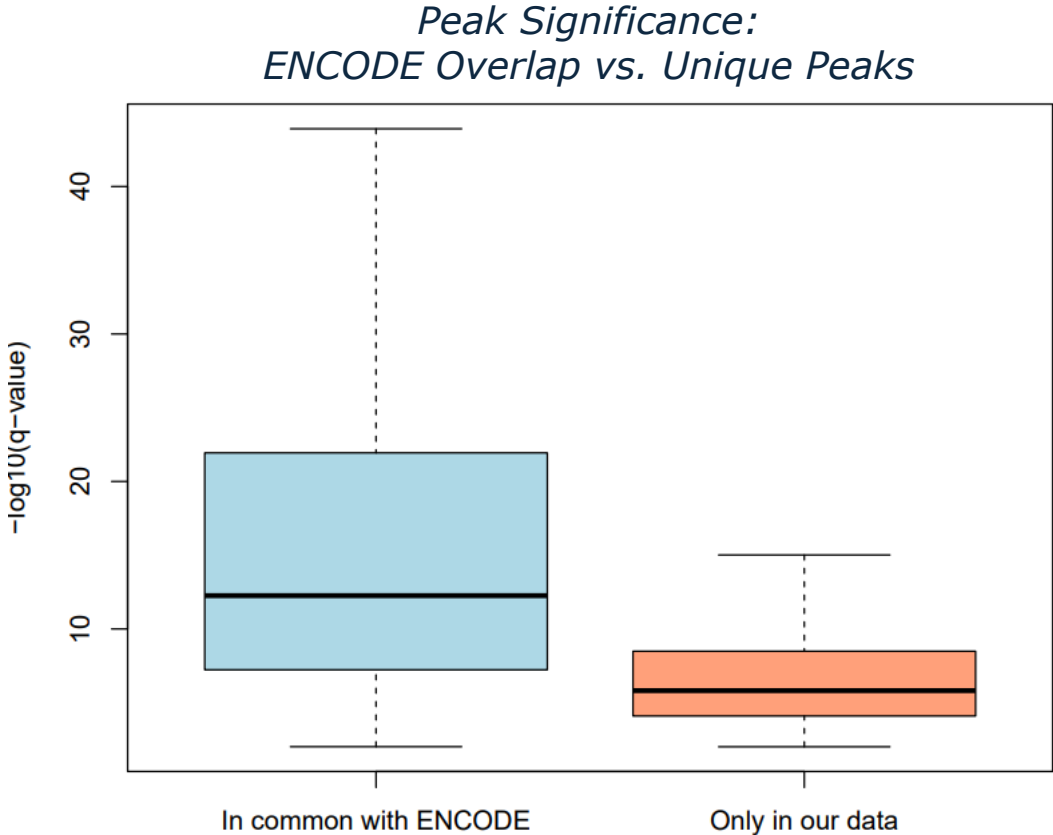
ENCODE comparison

Data	Simple Region Overlap	Jaccard Index
Replicate 1 vs ENCODE	19316	0.312
Replicate 2 vs ENCODE	17664	0.357
Intersection vs ENCODE	16521	0.335
Merged set vs ENCODE	17577	0.346

0.346
Highest Jaccard Index
(Intersection vs. Merged)
The **Merged set** was selected
as the final dataset.

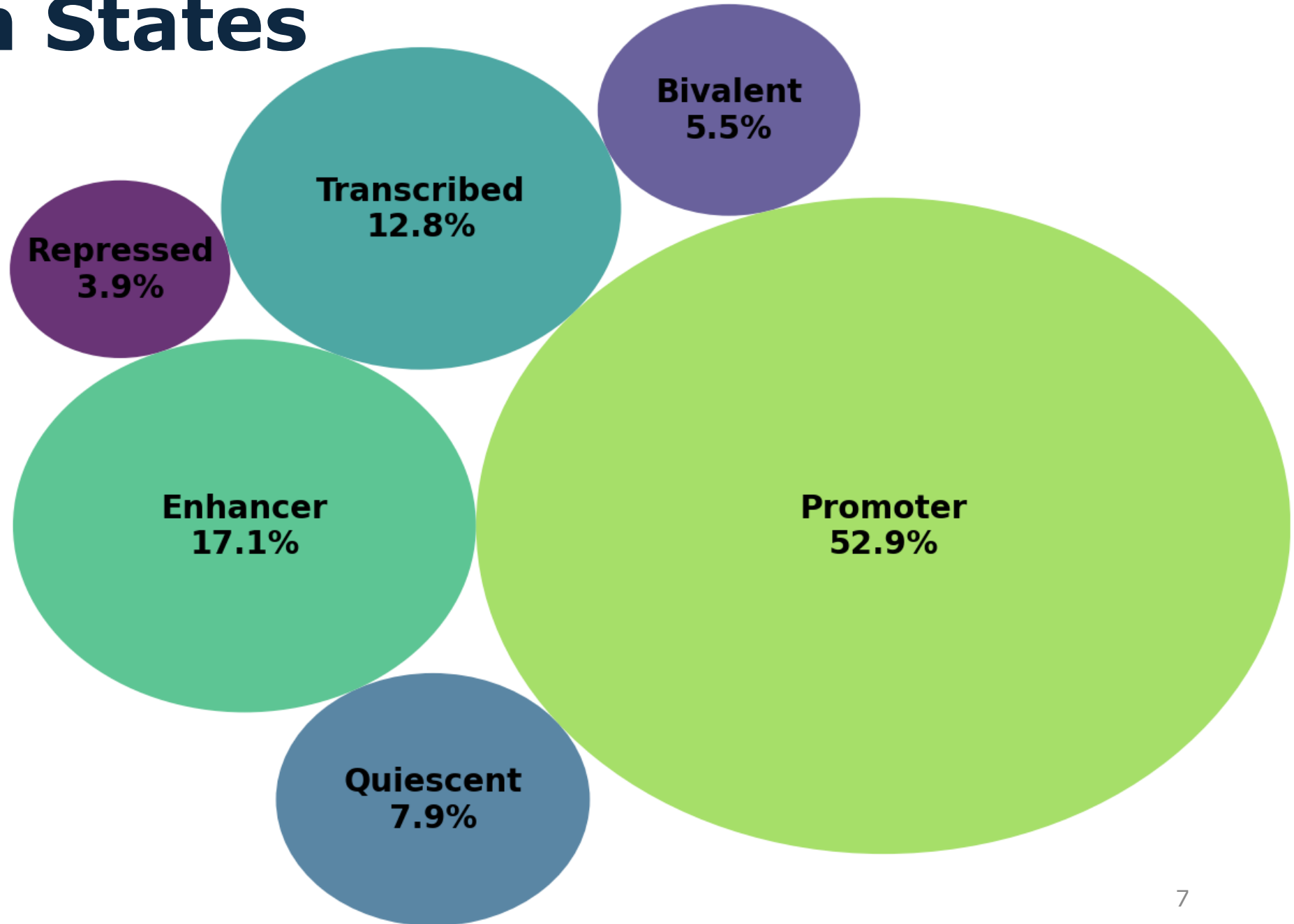
Blacklist Filtering

Regions removed by ENCODE Blacklist: **11**
→ Final clean summits: **26,297**



Chromatin States

*Distribution of
summits across
chromatin states.*

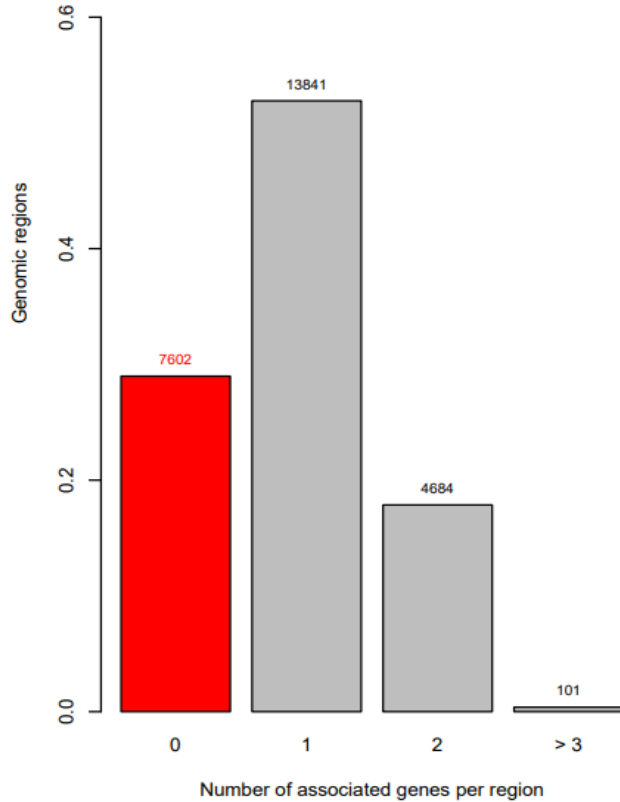


Target Genes (GREAT)

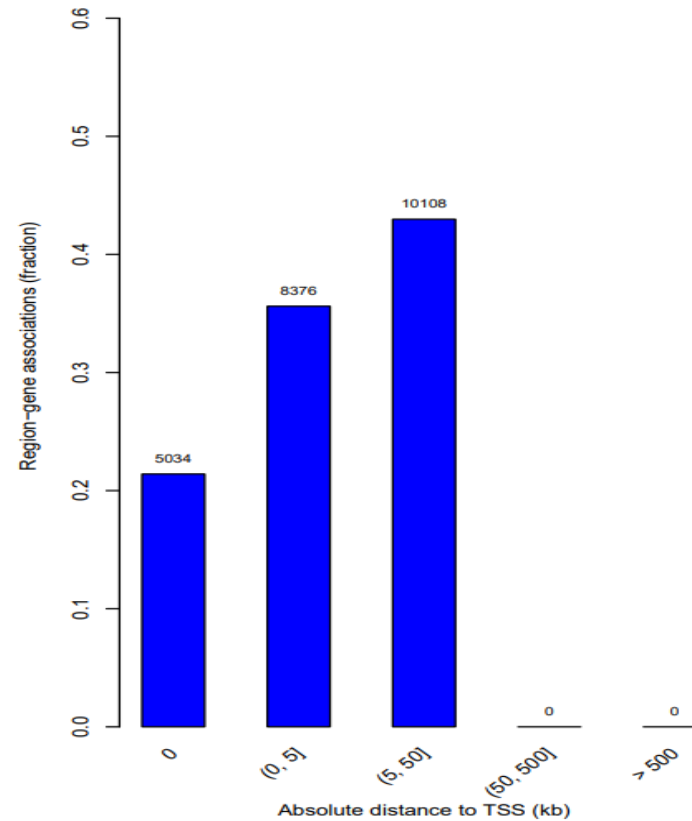
Target Genes Associated:

Promoter-bound (± 1 kb): 5,739 genes

Distal-bound (1 - 30 kb): 3,678 genes



Unassigned Peaks: 29%
Located at >30 kb from any TSS

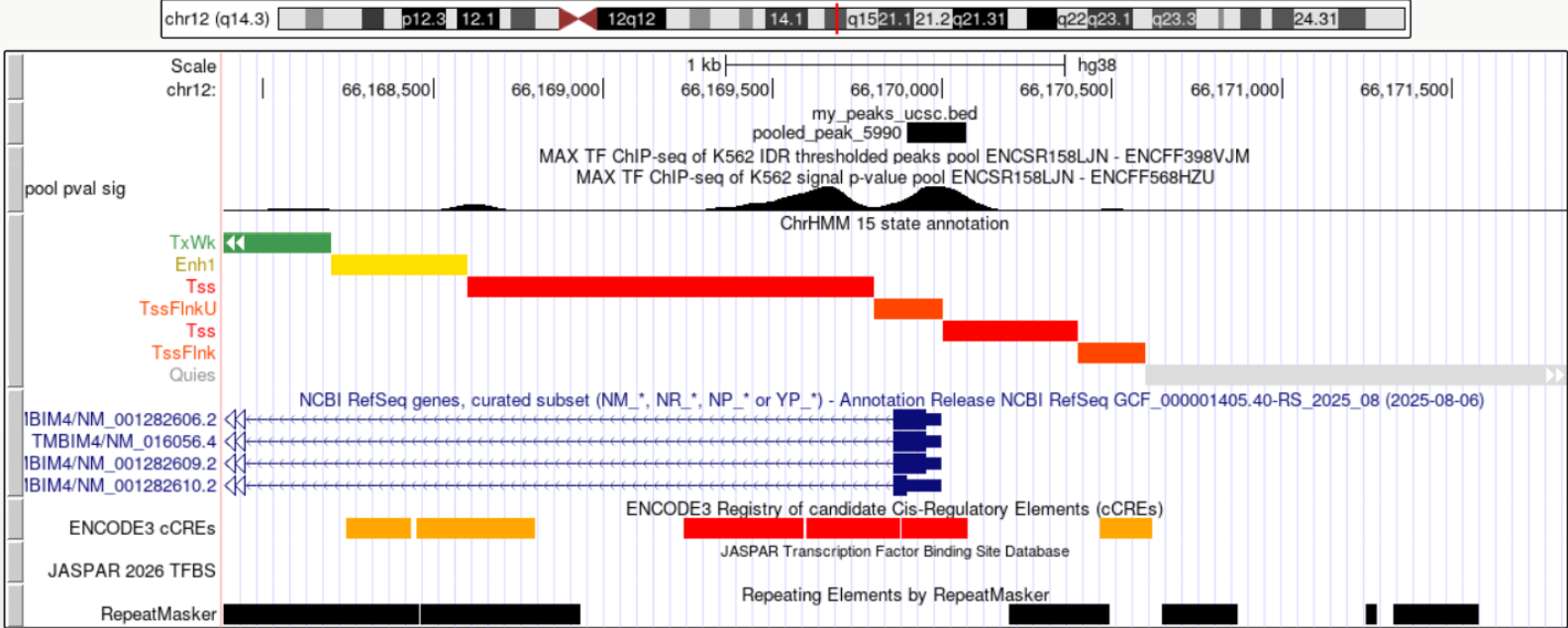


MAX heavily regulates itself!

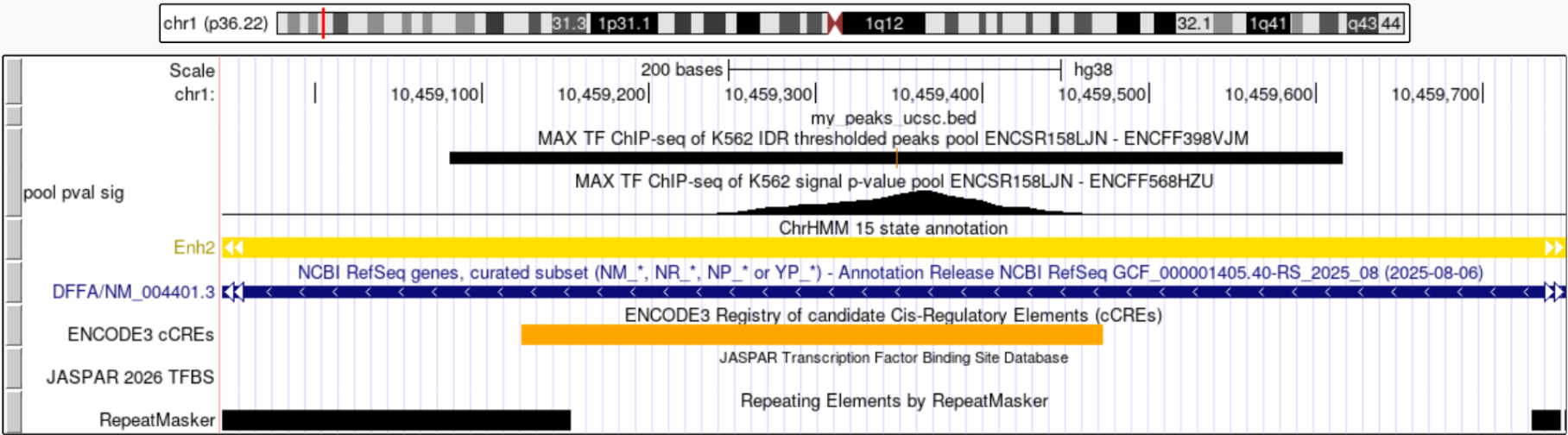
Two binding events on *chr14*:

- Directly at the Promoter (Distance: 0 bp)
- At a distal regulatory element (-12.9 kb)

Unique Peaks: UCSC browser view

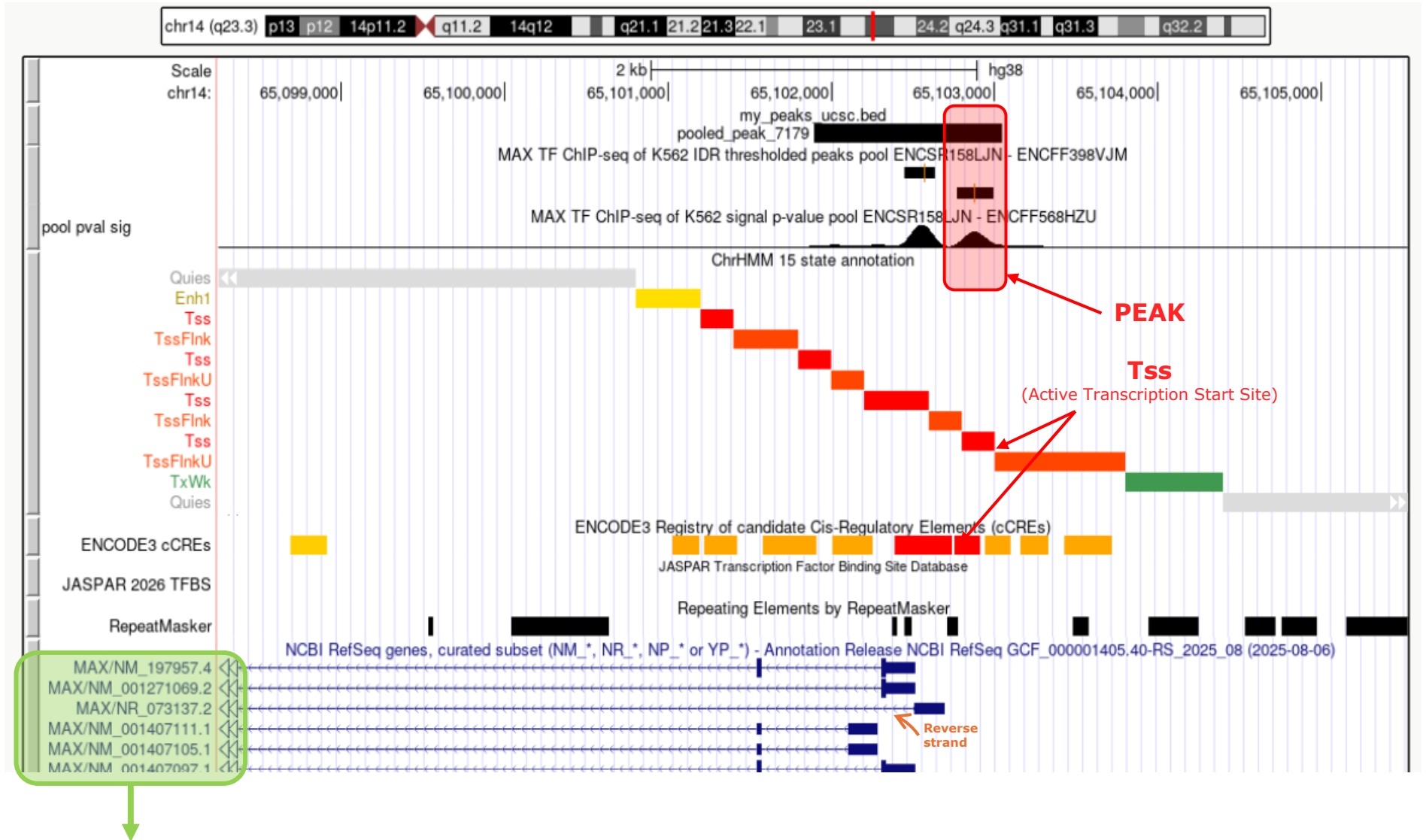


A peak found by me and not by ENCODE
chr12:66,168,000-66,171,800



A peak found by ENCODE and not by me: *false negative*
chr1:10,458,945-10,459,750

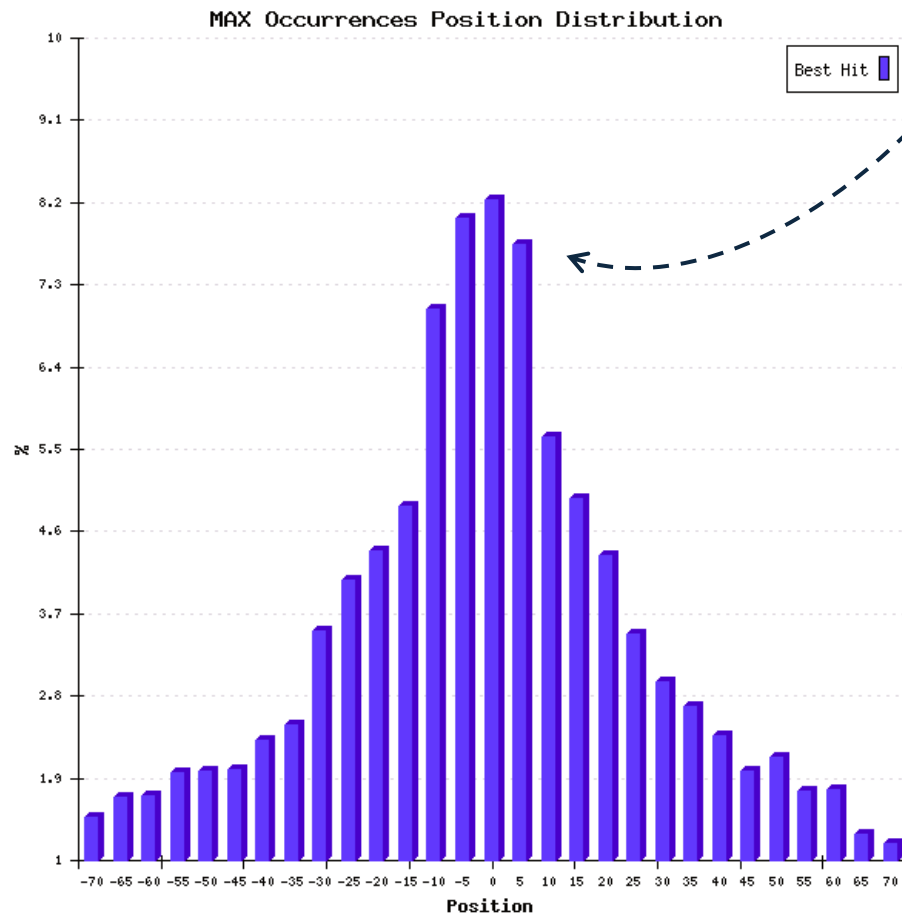
Evidence of MAX Auto-Regulation



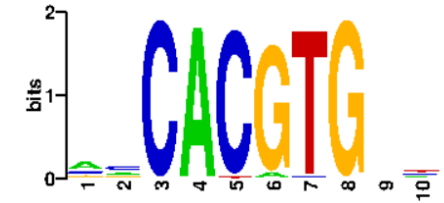
Multiple MAX alternative transcripts

Motif Analysis (PscanChIP)

At the very top of the enrichment list are the **MAX** motif and its heterodimer **MAX::MYC**.



The **MAX** binding motif is enriched directly at the peak summits.



Most Enriched Non-bHLH Motif: Creb3l2

- **Domain:** bZIP (basic leucine zipper)
- **Significance:** Lacks the helix-loop-helix structure but shows maximum enrichment (P-value: 0 OVER).
- **Cooperative Binding:** Known to co-occupy active promoters alongside MYC/MAX complexes.

References

- *Nair, S. K., & Burley, S. K. (2003). X-ray structures of Myc-Max and Mad-Max recognizing DNA. Cell, 112(2), 193-205.*
- *Blackwood, E. M., & Eisenman, R. N. (1991). Max: a helix-loop-helix zipper protein that forms a sequence-specific DNA-binding complex with Myc. Science, 251(4998), 1211-1217.*
- *Grandori, C., et al. (2000). The Myc/Max/Mad network and the transcriptional control of cell behavior. Annual review of cell and developmental biology, 16(1), 653-699.*